

4 (a) Find $\mathbf{M}^{-1}$, where $\mathbf{M} = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$. [1]

(b) Hence find, in terms of the constant k , the point of intersection of the planes

$$\begin{aligned} x + 2y + 3z &= 19, \\ -x + y + 2z &= 4, \\ -2x + y + 2z &= k. \end{aligned}$$

[3]

(c) In this question you must show detailed reasoning.

Find the acute angle between the planes $x + 2y + 3z = 19$ and $-x + y + 2z = 4$. [4]